

Temporal PhilD: Interpretation of Atoms and Metrics

Overview

This document explains how to interpret the 16 PhilD atoms and derived metrics when using **Takens delay embedding** on a single time series. The standard PhilD framework assumes two distinct signals (X, Y), but our temporal approach embeds one signal at 4 regularly-spaced time points.

1. The Temporal Embedding

Standard PhilD (Two Signals)

Source 1 (X): X_past → X_future
Source 2 (Y): Y_past → Y_future

Temporal Takens PhilD (Single Signal)

signal x(t) embedded as:

p1 = x(t) → "X_past"
p2 = x(t+τ) → "Y_past"
t1 = x(t+2τ) → "X_future"
t2 = x(t+3τ) → "Y_future"

Timeline: t —τ— t+τ —τ— t+2τ —τ— t+3τ
p1 p2 t1 t2

Key insight: "X" and "Y" are not independent signals—they are **phase-shifted copies** of the same process at different points in the embedding window.

2. The 16 PhilD Atoms

Atom Notation

Each atom is denoted αβ where:

- First letter (α):** Information type in the SOURCE (past)
- Second letter (β):** Information type in the TARGET (future)

Letter	Meaning	In Temporal Context
r	Redundancy	Info shared by BOTH p1 and p2
x	Unique to "X"	Info unique to p1 (earliest point)
y	Unique to "Y"	Info unique to p2 (early-mid point)
s	Synergy	Info requiring BOTH p1 AND p2 together

The 4x4 Atom Grid

	→ r (Red)	→ x (Un1)	→ y (Un2)	→ s (Syn)
r →	rtr	rtx	rty	rts
x →	xtr	xtx	xyx	xts
y →	ytr	ytx	yty	yts
s →	str	stx	sty	sts

Temporal Interpretation of Each Atom

Diagonal Atoms (Self-Transfer)

Atom	Standard Meaning	Temporal Meaning
rtr	Redundant info stays redundant	Persistent memory: Info stable across ALL 4 timepoints
xtx	X-unique info stays X-unique	Local persistence at $t \rightarrow t+2\tau$: Position 0 predicts position 2
yty	Y-unique info stays Y-unique	Local persistence at $t+\tau \rightarrow t+3\tau$: Position 1 predicts position 3
sts	Synergy creates synergy	Temporal integration: Info only decodable with ALL 4 points

Transfer Atoms (Cross-Position)

Atom	Standard Meaning	Temporal Meaning
xy	X-unique becomes Y-unique	Odd-even phase coupling: $t \rightarrow t+2\tau$ info shifts to $t+\tau \rightarrow t+3\tau$
yx	Y-unique becomes X-unique	Even-odd phase coupling: $t+\tau \rightarrow t+3\tau$ info shifts to $t \rightarrow t+2\tau$
xtr	X-unique becomes redundant	Info spreading: Early-unique info becomes shared
ytr	Y-unique becomes redundant	Info spreading: Mid-unique info becomes shared

Erasure Atoms (Information Loss)

Atom	Standard Meaning	Temporal Meaning
rtx	Redundant becomes X-unique	Memory decay (partial): Shared info lost at position 1,3
rt	Redundant becomes Y-unique	Memory decay (partial): Shared info lost at position 0,2
rts	Redundant becomes synergistic	Complexity emergence: Shared info becomes entangled

Causation Atoms (Up/Down)

Atom	Standard Meaning	Temporal Meaning
xts	X-unique creates synergy	Upward causation: Early info combines with other to create emergence
ys	Y-unique creates synergy	Upward causation: Mid info combines to create emergence
stx	Synergy becomes X-unique	Downward causation: Whole constrains early positions
sty	Synergy becomes Y-unique	Downward causation: Whole constrains mid positions
str	Synergy becomes redundant	Downward spread: Emergent info becomes persistent

3. High-Order Metrics: DYNAMICS_GROUPS

Storage: ['rtr', 'xtx', 'yty', 'sts']

Standard: Information preserved over time in both X and Y

Temporal interpretation:

- **Total temporal persistence** across the embedding window
- Measures how predictable the signal is from ANY earlier point to ANY later point
- High storage = strong autocorrelation structure at timescale τ

Processes with high Storage:

- COPY: $x[t] = x[t-1]$
 - AR(1) with high ϕ
 - Slowly varying signals
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Copy: ['xtx', 'yty']

Standard: Self-continuity within X and within Y

Temporal interpretation:

- **Autocorrelation at lag 2τ** (skip-one correlation)
- xtx : How well does $x(t)$ predict $x(t+2\tau)$?
- yty : How well does $x(t+\tau)$ predict $x(t+3\tau)$?

Important: This is NOT "copying" in the colloquial sense—it measures correlation at **double** the embedding lag.

Processes with high Copy:

- Signals with strong 2τ autocorrelation
 - Oscillations with period $\sim 4\tau$ (quarter-period spacing lands on same phase)
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Transfer: ['xty', 'ytx']

Standard: Information flow between $X \rightarrow Y$ and $Y \rightarrow X$

Temporal interpretation:

- **Phase relationship between embedding positions**
- xty : Info at $(t, t+2\tau)$ that predicts $(t+\tau, t+3\tau)$ pattern
- ytx : Info at $(t+\tau, t+3\tau)$ that predicts $(t, t+2\tau)$ pattern
- Measures "cross-talk" between even/odd positions

NOT true transfer entropy in the temporal case—there's no separate "source" and "target" process.

Processes with high Transfer:

- Oscillations ($\pi/2$ phase offset between even/odd positions)
 - AR(2) processes
 - Signals with specific phase relationships
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Erasure: ['rtx', 'rty']

Standard: Redundant information that becomes unique (lost from one stream)

Temporal interpretation:

- **Memory decay within the embedding window**
- Info that was shared at $(t, t+\tau)$ but only persists to one of $(t+2\tau, t+3\tau)$
- Indicates **partial forgetting** on timescale $\sim 2\tau$

Processes with high Erasure:

- Signals with mixed persistence (some components decay faster)
 - Transitions between states
 - Edge effects in windows
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Upward_causation: ['xts', 'yts', 'rts']

Standard: Parts (unique/redundant) create synergistic whole

Temporal interpretation:

- **Emergence of temporal patterns**
- Information from individual timepoints that **only becomes meaningful** when combined
- Measures how local dynamics create global structure

Processes with high Upward causation:

- XOR-like processes: $x[t] = x[t-1] \oplus x[t-2]$

- Systems where multi-scale interactions matter
- Nonlinear processes with delayed dependencies

Downward_causation: ['stx', 'sty', 'str']

Standard: Synergistic whole constrains the parts

Temporal interpretation:

- **Constraint by global temporal structure**
- Future synergistic patterns that **constrain** what can appear at individual timepoints
- Measures how the overall trajectory limits local variability

Processes with high Downward causation:

- Strongly oscillatory signals (global rhythm constrains local values)
- Attractor dynamics
- Signals with strong temporal structure

4. High-Order Metrics: IIT_METRICS

These are inspired by Integrated Information Theory (Mediano et al.) and the goofi PhilD implementation.

Information_storage: ['xtx', 'yty', 'rtr', 'sts']

Same as Storage in DYNAMICS_GROUPS.

Temporal interpretation: Total information preserved across the embedding window.

Transfer_entropy: ['xty', 'xtr', 'str', 'sty']

Standard: Information transferred between processes

Temporal interpretation:

- **Cross-position information flow** within the embedding
- NOT equivalent to true transfer entropy (which requires separate source/target)
- Measures "off-diagonal" flow: how info at one phase predicts another phase

Components:

- **xty** : Even→Odd phase coupling
- **xtr** : Early-unique spreading to redundant
- **str** : Synergy becoming redundant
- **sty** : Synergy becoming Y-unique

Warning: In temporal PhilD, this does NOT measure causal influence between systems—it measures internal phase relationships.

Causal_density: ['xtr', 'ytr', 'sty', 'str', 'str', 'xty', 'ytx', 'stx']

Standard: Complexity of causal interactions (note: **str** appears twice in original definition)

Temporal interpretation:

- **Mixing of information types** across the embedding
- High causal density = info doesn't stay in neat categories
- Measures how much information "moves around" between redundancy, unique, and synergy

Processes with high Causal density:

- Complex/chaotic dynamics
- Signals with multiple interacting timescales
- Non-stationary processes

Integrated_information: ['rts', 'xts', 'sts', 'sty', 'str', 'yts', 'ytx', 'stx', 'xty']

Standard: Φ -like measure: synergistic integration beyond the parts

Temporal interpretation:

- **Net temporal integration** across the embedding window
- In the code, **rtr** is subtracted: `integrated = sum(atoms) - rtr`
- This removes "trivial" redundancy to measure true emergence

- Components (all involve synergy):
- Synergy-producing: `rts` , `xts` , `yts` (upward causation)
 - Synergy-maintaining: `sts`
 - Synergy-distributing: `sty` , `str` , `stx` (downward causation)
 - Transfer: `ytx` , `xy`

- Processes with high Integrated information:
- XOR processes (pure synergy)
 - Complex temporal dependencies
 - Signals where knowing one timepoint tells you nothing without others

5. Summary Table: What Each Metric Actually Measures

Metric	Standard Interpretation	Temporal Takens Interpretation
Storage	Info preserved in X and Y	Total autocorrelation structure at scale τ
Copy	Self-continuity	Autocorrelation at lag 2τ
Transfer	$X \leftrightarrow Y$ flow	Phase coupling between even/odd embedding positions
Erasure	Info loss from shared pool	Partial memory decay within window
Upward	Parts $\rightarrow$ Whole	Local dynamics creating global patterns
Downward	Whole $\rightarrow$ Parts	Global structure constraining local dynamics
Integrated_info	Φ (synergistic integration)	True temporal emergence minus trivial persistence

6. Key Differences from Standard PhiID

1. **No independent sources:** X and Y are the SAME process, so "transfer" measures internal phase relationships, not cross-system influence.
2. **Autocorrelation dominates:** Redundancy (`r*` atoms) primarily reflects temporal autocorrelation, not shared information between independent sources.
3. **Synergy = temporal integration:** High `s*` atoms indicate information that requires MULTIPLE timepoints to decode—true temporal binding.
4. **τ is everything:** The embedding lag τ determines what timescale you're probing. Different τ values probe different temporal structures.
5. **"Copy" $\neq$ copying:** The Copy metric measures correlation at 2τ , not moment-to-moment persistence.

7. Practical Interpretation Guidelines

- When TII is HIGH (near 1):
- Same information dynamics at all τ values
 - Scale-invariant temporal structure
 - Consistent memory/integration across timescales

- When TII is LOW (near 0):
- Different dynamics at different τ values
 - Multi-scale processing
 - Timescale-specific information structure

- High Storage + Low Synergy:
- Simple, persistent signal
 - Strong autocorrelation
 - Example: Random walk, slow drift

Low Storage + High Synergy:

- Complex, integrated dynamics
- Information requires multiple timepoints
- Example: XOR process, chaotic systems

High Transfer:

- Strong phase relationships within embedding
- Oscillatory structure
- Example: Sine wave at appropriate τ

High Erasure:

- Memory decay within the window
- Transient dynamics
- Example: State transitions, non-stationary signals

8. References

- Mediano, P. A., et al. (2021). Towards an extended taxonomy of information dynamics via Integrated Information Decomposition.
- Rosas, F. E., et al. (2020). Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data.
- Lizier, J. T. (2012). JIDT: An information-theoretic toolkit for studying the dynamics of complex systems.

Appendix: Atom Quick Reference

Atom	Source → Target	Temporal Meaning
rtr	Redundant → Redundant	Persistent across ALL 4 points
rtx	Redundant → X-unique	Memory decay (lost at pos 1,3)
rty	Redundant → Y-unique	Memory decay (lost at pos 0,2)
rts	Redundant → Synergy	Shared info becomes entangled
xtr	X-unique → Redundant	Early info spreads to all
xtx	X-unique → X-unique	Persistence at lag 2τ (pos 0→2)
xty	X-unique → Y-unique	Even→Odd phase coupling
xts	X-unique → Synergy	Early info creates emergence
ytr	Y-unique → Redundant	Mid info spreads to all
ytx	Y-unique → X-unique	Odd→Even phase coupling
yty	Y-unique → Y-unique	Persistence at lag 2τ (pos 1→3)
yts	Y-unique → Synergy	Mid info creates emergence
str	Synergy → Redundant	Emergent info becomes persistent
stx	Synergy → X-unique	Whole constrains even positions
sty	Synergy → Y-unique	Whole constrains odd positions
sts	Synergy → Synergy	Pure temporal integration